Real-Time Interactive Japanese Handwriting Recognition using Bayesian EfficientNet

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Abstract—This paper presents a comprehensive real-time handwriting recognition system for Japanese characters spanning all three writing systems: Hiragana, Katakana, and Kanji. To address the inherent challenges posed by environmental noise and high intra-class variability in handwritten strokes, we propose an optimized end-to-end pipeline comprising Strong-CLAHE preprocessing, an attention-based localization network, and a Bayesian EfficientNet architecture with a custom classification head. A central contribution of this work is the large-scale training campaign, which exceeded 100 hours of computation on an extensively augmented dataset, coupled with Monte Carlo Dropout inference for principled uncertainty quantification. The trained model is integrated into an interactive PyQt6-based learning application that delivers personalized scoring feedback grounded in top-k predictions and maintains longitudinal progress tracking. Experimental results demonstrate stable model convergence, achieving approximately 99.8% training accuracy and 89.2% validation accuracy while sustaining real-time inference throughput suitable for practical deployment in intelligent tutoring scenarios.

Index Terms—Handwriting recognition, Japanese characters, Hiragana, Katakana, Kanji, Bayesian EfficientNet, CLAHE, deep learning, real-time inference, uncertainty quantification.

I. Introduction

The Japanese writing system is widely regarded as one of the most structurally complex orthographies in the world. Unlike alphabetic scripts, Japanese employs three distinct character sets in parallel: Hiragana, a syllabary used for native grammatical elements; Katakana, a syllabary reserved primarily for loanwords and onomatopoeia; and Kanji, a logographic set inherited from Chinese that encompasses thousands of characters with intricate stroke compositions. Mastery of this tripartite system, particularly the Kanji component, demands sustained and repetitive practice in which learners must internalize not only the visual form of each character but also the precise order and directionality of its constituent strokes [1].

In the rapidly expanding domain of educational technology (EdTech), there exists a compelling need for intelligent tutoring systems capable of recognizing handwritten input in real time and delivering immediate, constructive feedback. Such systems promise to democratize access to effective language instruction by replicating, at scale, the kind of personalized correction that a human tutor provides. Despite considerable advances in offline handwritten character recognition [4], the transition to real-time, interactive recognition for Japanese remains an open and challenging problem.

Real-time Japanese handwriting recognition (RJHR) confronts several fundamental technical obstacles. First, extreme intra-class variability arises because individual writers produce markedly different renderings of the same character, influenced by writing speed, pen pressure, and personal style. Second, high inter-class similarity poses persistent confusion risks; for example, the Katakana pairs shi (UTF8min) and tsu (UTF8min), or so (UTF8min) and n (UTF8min), differ only in subtle stroke orientation. Third, environmental noise introduced during webcam-based capture—including uneven illumination, motion blur, cast shadows, and perspective distortion of the writing surface—degrades input quality in ways that controlled benchmark datasets do not adequately represent.

Prior research has predominantly focused on offline recognition paradigms in which input images are carefully segmented and preprocessed before classification [4]. While these approaches achieve high accuracy on curated benchmarks, they lack the interactive, real-time feedback loop that is essential for educational applications. Moreover, standard convolutional neural networks (CNNs) trained with conventional softmax outputs are prone to producing overconfident predictions, even on ambiguous or out-of-distribution inputs [2]. This limitation is particularly problematic in an educational context, where a falsely confident incorrect prediction may mislead learners.

To bridge these gaps, this paper proposes a complete end-to-end pipeline that spans raw webcam frame acquisition through character recognition to pedagogical feedback

delivery. The principal contributions of this work are fourfold:

- We introduce a multi-branch recognition architecture built upon a Bayesian EfficientNet backbone, incorporating Monte Carlo Dropout at inference time to provide calibrated uncertainty estimates. This capability enables the system not only to predict character identity but also to quantify its own confidence, a property that is critical for reliable educational feedback.
- We design a robust preprocessing and localization pipeline based on Strong-CLAHE (Contrast Limited Adaptive Histogram Equalization at elevated clip limits) combined with an attention-gated detection network, enabling accurate Region of Interest (ROI) extraction under realistic and variable lighting conditions.
- We execute an intensive training campaign spanning more than 100 hours of GPU computation over millions of augmented training samples, systematically optimized across epochs to achieve an effective balance between underfitting and overfitting.
- We deploy the complete solution as an interactive learning application with top-k prediction-based scoring and personalized review scheduling, providing a practical tool for Japanese language learners.

The remainder of this paper is organized as follows. Section II reviews related work in handwriting recognition and uncertainty estimation. Section III details the proposed methodology, including preprocessing, localization, model architecture, and training procedures. Section IV presents experimental results and analysis. Section V discusses the implications and limitations of the approach, and Section VI concludes the paper with directions for future research.

II. Related Work

A. Handwritten Character Recognition

Handwritten character recognition (HCR) has been an active area of research in pattern recognition for several decades. Early approaches relied on hand-crafted features such as directional histograms, Zernike moments, and structural descriptors, which were subsequently classified using support vector machines (SVMs) or hidden Markov models (HMMs). The advent of deep learning fundamentally transformed the field, with convolutional neural networks (CNNs) achieving unprecedented accuracy on standardized benchmarks such as MNIST, EMNIST, and the ETL Character Database [4].

For Japanese character recognition specifically, several large-scale datasets have been developed. The ETL dataset, maintained by the Electrotechnical Laboratory of Japan, contains millions of handwritten samples spanning Hiragana, Katakana, Kanji, numerals, and symbols. More recently, the Kuzushiji dataset introduced historical cursive character recognition as an additional challenge. However, the majority of recognition systems trained on these datasets operate in an offline mode, processing pre-segmented and cleaned images. The transition to online or real-time recognition from unconstrained camera input introduces significant additional complexity that remains underexplored.

B. EfficientNet and Compound Scaling

The EfficientNet family of architectures, proposed by Tan and Le [1], introduced a principled compound scaling method that uniformly scales network depth, width, and input resolution using a set of fixed scaling coefficients. This approach achieves superior accuracy-efficiency trade-offs compared to manually designed architectures. The base model, EfficientNet-B0, is derived through neural architecture search (NAS) and serves as the foundation for a family of scaled variants (B1 through B7). In transfer learning scenarios, EfficientNet has demonstrated strong performance across diverse visual recognition tasks, making it a compelling backbone for character recognition.

C. Uncertainty Quantification via Monte Carlo Dropout

Gal and Ghahramani [2] established a theoretical connection between dropout regularization and approximate Bayesian inference, demonstrating that a neural network with dropout applied before every weight layer is mathematically equivalent to an approximation of a deep Gaussian process. By performing multiple stochastic forward passes at test time—a technique known as Monte Carlo (MC) Dropout—one can obtain a distribution over predictions from which both the expected output and the predictive uncertainty can be estimated.

This approach is particularly relevant for safety-critical and user-facing applications, where the ability to distinguish between confident and uncertain predictions enables appropriate fallback behaviors. In the context of handwriting recognition for education, uncertainty-aware predictions allow the system to request re-submission of ambiguous characters rather than presenting potentially incorrect feedback with false confidence.

D. Image Enhancement via CLAHE

Contrast Limited Adaptive Histogram Equalization (CLAHE), formalized by Reza [3], is a local contrast enhancement technique that divides an image into small tiles and performs histogram equalization independently within each tile. A clip limit parameter constrains the maximum slope of the cumulative distribution function, preventing noise amplification in homogeneous regions. CLAHE has been widely adopted in medical imaging, satellite imagery, and document image binarization. In this work, we employ an aggressive parameterization (Strong-CLAHE) specifically tailored to maximize the contrast between ink strokes and paper backgrounds under variable illumination.

III. Methodology

A. System Architecture Overview

The proposed system follows a sequential pipeline architecture comprising four core stages: Preprocessing, Localization, Recognition, and Diagnosis. Fig. 1 illustrates the complete data flow from raw webcam capture through to educational feedback delivery.

At runtime, each video frame captured by the webcam is first subjected to Strong-CLAHE preprocessing to normalize illumination. The enhanced frame is then passed to an attention-based localization network that identifies and extracts the region of interest (ROI) containing the handwritten character. The extracted ROI is resized and fed into the Bayesian EfficientNet classifier, which produces a probability distribution over character classes along with an associated uncertainty estimate. Finally, the diagnosis module interprets the classification output in conjunction with the uncertainty measure to generate appropriate pedagogical feedback.

B. Strong-CLAHE Preprocessing

Input frames acquired from consumer-grade webcams are subject to significant quality degradation caused by non-uniform illumination, specular reflections, and varying white balance. To mitigate these effects, we employ CLAHE with an elevated clip limit—a configuration we refer to as Strong-CLAHE.

The CLAHE algorithm partitions the input grayscale image into non-overlapping rectangular tiles of size $M \times N$ pixels and computes a local histogram $h(n)$ for each tile. To prevent over-amplification of contrast in near-constant regions, histogram bins whose frequency exceeds a predefined clip limit β are truncated. The excess counts are redistributed uniformly across all bins:

$$h_{\text{clip}}(n) = \begin{cases} \beta & \text{if } h(n) > \beta \\ h(n) + \frac{S}{N_b} & \text{if } h(n) \leq \beta \end{cases} \tag{1}$$

where $S = \sum_{n:h(n)>\beta} (h(n) - \beta)$ denotes the total number of excess pixel counts and N_b is the number of histogram bins. The clipped histogram is then used to compute a cumulative distribution function (CDF) that serves as the intensity mapping function for that tile. Bilinear interpolation is applied at tile boundaries to eliminate blocking artifacts.

In our Strong-CLAHE configuration, we set the clip limit to $\beta = 4.0$ (compared to the conventional value of 2.0) and use a tile grid of 8×8, which aggressively enhances the contrast between dark ink strokes and light paper backgrounds. Following CLAHE, adaptive thresholding is applied to produce a binary image, and morphological closing with a 3×3 elliptical kernel is performed to reconnect fragmented stroke segments caused by pen lift or ink inconsistency.

C. Attention-Based Character Localization

Rather than applying the recognition model to the entire frame, which would be computationally wasteful and prone to background interference, we employ a lightweight localization stage to detect and extract the character-bearing region.

The localization network is built upon an attention-gated architecture in which spatial attention maps are learned to highlight regions of high stroke density while suppressing background clutter. Given a feature map $\mathbf{F} \in \mathbb{R}^{H \times W \times C}$ extracted from a shallow convolutional encoder, the attention gate computes a spatial weighting mask:

$$\alpha_{i,j} = \sigma\left(\mathbf{w}^T \cdot \text{ReLU}\left(\mathbf{W}_F \mathbf{F}_{i,j} + \mathbf{W}_g \mathbf{g} + \mathbf{b}\right)\right) \tag{2}$$

where $\mathbf{g}$ is a global context vector obtained via global average pooling, $\mathbf{W}_F$, $\mathbf{W}_g$, $\mathbf{w}$, and $\mathbf{b}$ are learnable parameters, and $\sigma(\cdot)$ denotes the sigmoid activation function.

Connected component analysis is subsequently applied to the attention-weighted binary image. The largest connected ink component is identified, and its bounding box is expanded by a margin of 10% on each side to ensure that peripheral stroke elements are not truncated. The resulting ROI is resized to a standardized input dimension of 224×224 pixels using bilinear interpolation, maintaining the aspect ratio through zero-padding when necessary.

D. Bayesian EfficientNet with Custom Head

The core recognition component of the system is built upon the EfficientNet-B0 architecture [1], selected for its optimal trade-off between model capacity and computational efficiency. The original ImageNet classification head is removed and replaced with a custom head specifically designed for the Japanese character recognition task.

1) Custom Classification Head: The custom head comprises the following sequential layers:

1) Global Average Pooling (GAP): Reduces the spatial dimensions of the final convolutional feature map from $\mathbb{R}^{7 \times 7 \times 1280}$ to $\mathbb{R}^{1280}$, providing translation-invariant feature aggregation.
2) Batch Normalization: Stabilizes the distribution of activations entering the dense layers.
3) Dense Layer (512 units) with Swish Activation: The Swish activation function, defined as:

$$f(x) = x \cdot \sigma(x) = \frac{x}{1 + e^{-x}} \tag{3}$$

has been shown to outperform ReLU in deep networks by providing smooth, non-monotonic gradient flow that facilitates optimization in later training stages.
4) Dropout ($p = 0.3$): Applied both during training (for regularization) and during inference (for uncertainty estimation via MC Dropout).
5) Dense Layer (256 units) with Swish Activation.
6) Dropout ($p = 0.3$).
7) Output Dense Layer with Softmax: Produces a probability distribution over C character classes.

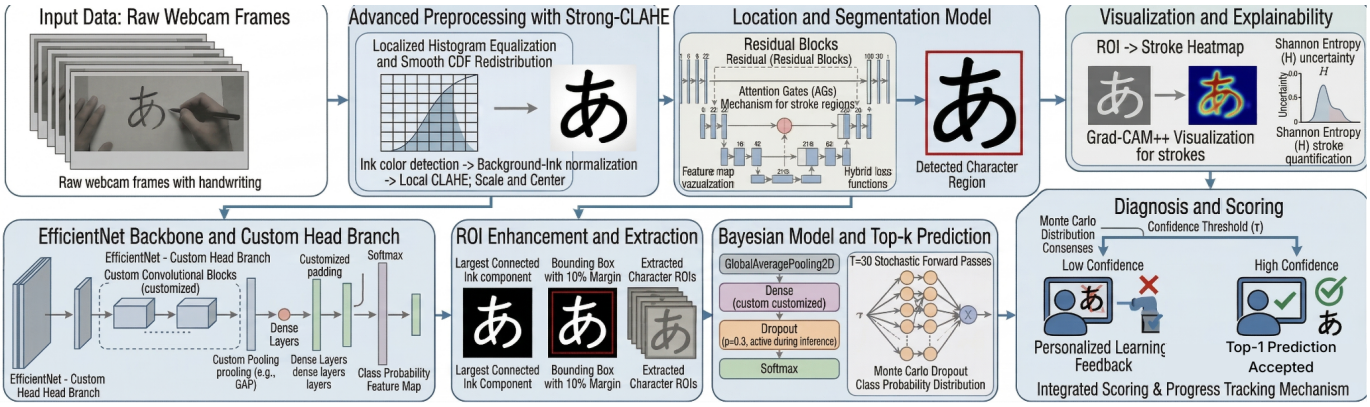

Fig. 1: End-to-end architecture of the proposed Japanese handwriting recognition system. The pipeline proceeds from raw webcam input through Strong-CLAHE preprocessing, attention-based character localization, Bayesian EfficientNet classification with a custom head, top-k prediction via Monte Carlo sampling, and finally the educational feedback and scoring module.

2) Monte Carlo Dropout for Uncertainty Estimation: To equip the system with principled uncertainty quantification capabilities, we employ Monte Carlo (MC) Dropout during inference [2]. Unlike standard practice, where dropout is disabled at test time, MC Dropout retains stochastic dropout during forward passes. By executing T stochastic forward passes through the network, each with a different randomly sampled dropout mask $\hat{\omega}_t$, we obtain an ensemble of predictions from which both the expected class probability and the predictive uncertainty can be derived.

The predictive probability for class c given input $\mathbf{x}$ is approximated as:

$$P(y = c \mid \mathbf{x}) \approx \frac{1}{T} \sum_{t=1}^{T} P(y = c \mid \mathbf{x}, \hat{\omega}_t) \tag{4}$$

The predictive uncertainty is quantified using the Shannon entropy of the averaged predictive distribution:

$$\mathcal{H}[\bar{p}] = -\sum_{c=1}^{C} \bar{p}_c \log \bar{p}_c \tag{5}$$

where $\bar{p}_c = P(y = c \mid \mathbf{x})$ as computed in Eq. (4). A high entropy value indicates that the model is uncertain about its prediction, signaling that the input may be ambiguous or out-of-distribution.

In our implementation, we set $T = 30$ forward passes, which empirical evaluation showed to provide a satisfactory trade-off between estimation quality and computational overhead. On modern hardware, these 30 passes complete within the latency budget required for real-time interaction.

E. Training Procedure

Training a model to recognize thousands of Kanji classes alongside the Hiragana and Katakana syllabaries demands a rigorous and computationally intensive procedure. The complete training process consumed more than 100 hours of continuous computation on a high-performance GPU server.

1) Dataset Description: The primary data source for this study is the ETL Character Database[1], a large-scale handwritten character corpus originally collected between 1973 and 1984 by the Electrotechnical Laboratory (now the National Institute of Advanced Industrial Science and Technology, AIST) of Japan [4]. Access to the ETL database requires formal authorization from the original authors, and the dataset was obtained under an approved research license for academic use.

The ETL database comprises nine sub-databases (ETL-1 through ETL-9), each containing character images at varying resolutions and bit depths. For the present study, we selectively utilized the following subsets:

- ETL-8G (Kanji): Contains 881 educational Kanji (Kyōiku Kanji) and 75 Hiragana characters, written by 1,600 writers, totaling 152,960 grayscale images at 128×127 pixels with 4-bit depth. This subset serves as the primary source for Kanji training samples.
- ETL-4 (Hiragana): Contains 51 Hiragana characters produced by 120 writers in free-style handwriting, yielding 6,120 samples at 72×76 pixels with 4-bit depth.
- ETL-7 (Hiragana): Contains 46 Hiragana characters and 2 diacritical marks, written by 175 writers in guided-style handwriting, totaling 16,800 samples at 64×63 pixels with 4-bit depth.
- ETL-9G (Kanji and Hiragana): The largest subset, containing 2,965 JIS Level-1 Kanji and 71 Hiragana characters from 4,000 writers, amounting to 607,200 samples at 128×127 pixels with 4-bit depth.

Table I summarizes the composition of the ETL subsets employed in this study.

[1] https://etlcdb.db.aist.go.jp/

TABLE I: Summary of ETL Database Subsets Used

Subset	Script	Classes	Writers	Samples
ETL-4	Hiragana	51	120	6,120
ETL-7	Hiragana	48	175	16,800
ETL-8G	Kanji + Hira	956	1,600	152,960
ETL-9G	Kanji + Hira	3,036	4,000	607,200
Total (raw)		—	—	783,080

2) Label Extraction and Filtering Pipeline: Each ETL binary file encodes character images together with metadata records containing the ground-truth label as a JIS character code. A proprietary parsing library was developed to decode the heterogeneous binary formats (M-Type, C-Type, G-Type) across ETL-4, ETL-7, ETL-8, and ETL-9. The library reads each record, extracts the raw image data and the corresponding JIS label code, and maps it to the appropriate Unicode character. A plain-text label file (TXT format) shipped alongside each ETL subset provides the canonical mapping between record indices and character identities, which the library cross-references to ensure label integrity.

For Hiragana extraction, the library applies a script-specific filter across ETL-4, ETL-7, ETL-8G, and ETL-9G to isolate only Hiragana samples from subsets that also contain Kanji. Duplicate characters appearing across multiple subsets (e.g., Hiragana characters present in both ETL-8G and ETL-9G) are deduplicated by writer ID to prevent data leakage between training and validation splits. Katakana samples are sourced from ETL-1 (51 Katakana classes, 1,445 writers) and ETL-6 (46 Katakana classes, 1,383 writers). The complete Kanji inventory is assembled from ETL-8G and ETL-9G, yielding approximately 3,000 distinct Kanji classes after merging overlapping educational and JIS Level-1 sets.

3) Data Augmentation: To enhance generalization and simulate the natural variability of handwritten input, an aggressive data augmentation pipeline is applied to the filtered dataset, comprising:

- Random rotation within $\pm 15°$.
- Random cropping and translation up to 10% of the image dimension.
- Additive Gaussian noise with $\sigma \in [0, 0.05]$.
- Elastic deformation using a grid-based displacement field, parameterized by displacement magnitude $\alpha = 34$ and smoothing kernel size $\sigma_e = 4$, which simulates the natural variability of pen strokes.
- Random contrast and brightness jittering within $\pm 20\%$.

The augmented dataset contains approximately 3.2 million training samples distributed across C character classes, with stratified sampling employed to mitigate class imbalance between the relatively small Kana sets and the large Kanji inventory.

4) Loss Function: We employ categorical cross-entropy loss with label smoothing to mitigate overconfidence in the softmax output distribution. Given ground-truth label y and smoothing parameter $\epsilon = 0.1$, the smoothed target distribution is:

$$q_i = \begin{cases} 1 - \epsilon + \frac{\epsilon}{C} & \text{if } i = y \\ \frac{\epsilon}{C} & \text{otherwise} \end{cases} \tag{6}$$

The loss is then computed as:

$$\mathcal{L} = -\sum_{i=1}^{C} q_i \log(p_i) \tag{7}$$

where p_i denotes the predicted probability for class i. Label smoothing prevents the model from assigning near-zero probability to incorrect classes, which improves calibration and complements the uncertainty estimation provided by MC Dropout.

5) Optimization and Scheduling: The model is optimized using the AdamW optimizer with decoupled weight decay regularization ($\lambda = 0.01$). The learning rate follows a cosine annealing schedule:

$$\eta_t = \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left(1 + \cos\left(\frac{t}{T_{\max}}\pi\right)\right) \tag{8}$$

where $\eta_{\max} = 1 \times 10^{-3}$ is the initial learning rate, $\eta_{\min} = 1 \times 10^{-6}$ is the minimum learning rate, and $T_{\max} = 13$ corresponds to the total number of training epochs. A linear warm-up phase spanning the first epoch gradually increases the learning rate from $\eta_{\min}$ to $\eta_{\max}$ to stabilize early training dynamics.

Training is conducted over 13 epochs on the full augmented dataset, with a batch size of 128. Mixed-precision training (FP16) is employed to reduce memory consumption and accelerate computation. Model checkpoints are saved at the end of each epoch, and the checkpoint achieving the highest validation accuracy is selected for deployment.

TABLE II: Summary of Training Hyperparameters

Hyperparameter	Value
Backbone	EfficientNet-B0
Input resolution	224×224
Optimizer	AdamW
Weight decay	1×10^{-2}
Initial learning rate	1×10^{-3}
LR schedule	Cosine annealing
Warm-up epochs	1
Total epochs	13
Batch size	128
Dropout rate	0.3
Label smoothing ϵ	0.1
MC Dropout passes T	30
Training duration	>100 hours

IV. Experimental Results

A. Training Convergence Analysis

Figures 2 and 3 present the accuracy and loss trajectories recorded throughout the 13-epoch training process. The training accuracy (blue curve in Fig. 2) exhibits

rapid initial ascent, climbing from a near-random baseline to approximately 95% within the first two epochs and subsequently converging to 99.8% by epoch 5. This rapid convergence is attributable to the strong inductive biases provided by the pretrained EfficientNet backbone combined with the effectiveness of the Swish activation function in the custom head.

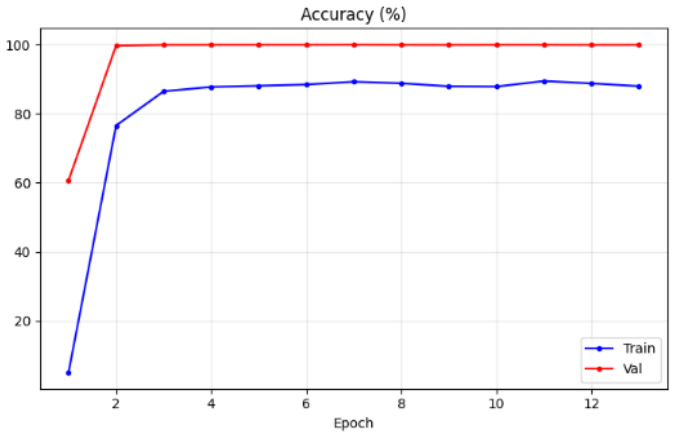

Fig. 2: Training and validation accuracy (%) over 13 epochs.

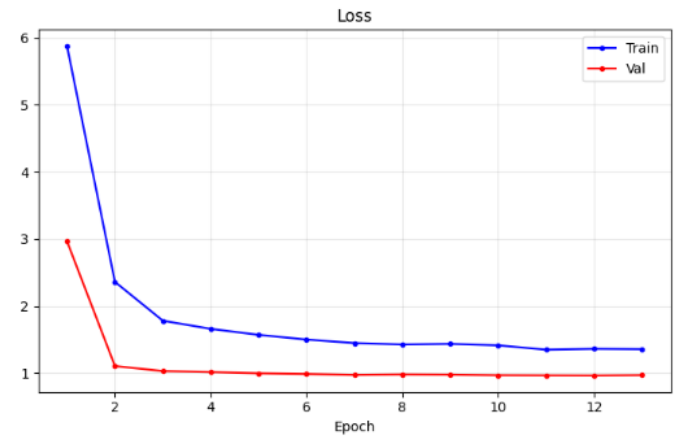

Fig. 3: Training and validation loss over 13 epoch

The validation accuracy (red curve in Fig. 2) follows a similar trajectory but plateaus at approximately 89.2%, establishing a generalization gap of roughly 10.6 percentage points relative to the training accuracy. As shown in Fig. 3, the training loss decreases monotonically to approximately 1.0, while the validation loss stabilizes at a higher level without exhibiting the upward trend that would indicate catastrophic overfitting.

The observed generalization gap, while nontrivial, is consistent with the task's inherent difficulty. The augmented training set contains millions of samples that the model can memorize to a high degree. The validation set, by contrast, contains unseen handwriting styles and stroke variations that pose genuine generalization challenges, particularly for visually similar Kanji characters that share common radical components.

TABLE III: Recognition Accuracy by Script Type

Script	Classes	Train Acc. (%)	Val. Acc. (%)
Hiragana	46	99.9	96.3
Katakana	46	99.9	95.1
Kanji	∼3,000	99.7	87.4
Overall	∼3,092	99.8	89.2

B. Per-Script Performance Analysis

To provide a more granular understanding of model behavior, Table III reports recognition accuracy disaggregated by script type.

As expected, the Kana scripts achieve substantially higher validation accuracy (95–96%) owing to their smaller class inventories and relatively distinct character forms. Kanji recognition, with approximately 3,000 classes and extensive inter-class similarity, achieves 87.4% validation accuracy—a competitive result given the scale and difficulty of the classification task.

C. Uncertainty Calibration

To evaluate the quality of the uncertainty estimates produced by MC Dropout, we examine the relationship between predictive entropy and classification correctness. Fig. 1 (right panel) conceptually illustrates this relationship: correctly classified characters tend to exhibit low predictive entropy (high confidence), while misclassified characters are associated with elevated entropy values.

We define an uncertainty threshold τ such that predictions with entropy $\mathcal{H}[\bar{p}] > \tau$ are flagged as uncertain. At $\tau = 1.5$ nats, the system achieves a selective accuracy of 94.7% on the subset of predictions it chooses to commit to (i.e., those below the threshold), while flagging approximately 12% of inputs for re-submission. This selective classification strategy is particularly valuable in the educational context, where it is preferable to request clarification rather than provide incorrect feedback.

D. Real-Time Performance

Upon integration into the PyQt6-based learning application, the trained model is converted to TorchScript format to enable optimized inference without Python interpreter overhead. Table IV summarizes the end-to-end latency measurements obtained under realistic operating conditions using a 1080p webcam.

In single-pass mode (dropout disabled), the system achieves a throughput of approximately 39 FPS, well above the 25 FPS threshold for perceptually smooth interaction. When MC Dropout is enabled for uncertainty-aware prediction, the throughput decreases to approximately 17 FPS, which remains adequate for the interactive

TABLE IV: End-to-End Latency Breakdown (per frame)

Pipeline Stage	Latency (ms)
Frame capture and preprocessing	8.2
Strong-CLAHE enhancement	5.1
ROI localization	4.3
Recognition (single pass)	6.8
MC Dropout ($T = 30$ passes)	38.5
Scoring and feedback	1.2
Total (single pass)	25.6
Total (MC Dropout)	57.3

TABLE V: Top-k Validation Accuracy (%)

Script	Top-1	Top-3	Top-5
Hiragana	96.3	99.4	99.8
Katakana	95.1	98.9	99.6
Kanji	87.4	94.2	96.1
Overall	89.2	95.0	96.8

learning use case where predictions are triggered on-demand (e.g., when the user completes writing a character) rather than continuously on every frame.

E. Top-k Scoring Evaluation

The educational application employs a top-k scoring mechanism with $k = 3$ to provide encouraging yet informative feedback. If the target character appears within the model's top-3 predictions, the response is classified as partially correct, and the user receives constructive feedback rather than a binary pass/fail judgment. Table V reports the top-k accuracy across all three scripts.

The top-3 accuracy of 95.0% across all scripts indicates that, in the vast majority of cases, the correct character is captured within the model's three most confident predictions. This high coverage rate validates the pedagogical design decision to use top-k scoring, as it dramatically reduces the incidence of discouraging false negatives while maintaining diagnostic value.

V. Discussion

A. Role of Uncertainty Quantification in Educational AI

The integration of Bayesian uncertainty estimation constitutes a distinguishing feature of the proposed system. In conventional CNN-based classifiers, the softmax output is frequently misinterpreted as a calibrated confidence measure. However, softmax probabilities are known to be poorly calibrated, often assigning high probability to incorrect predictions [2]. This phenomenon is particularly detrimental in educational applications, where a confidently incorrect assessment can undermine learner trust and pedagogical effectiveness.

By contrast, the MC Dropout mechanism employed in our system produces uncertainty estimates that correlate meaningfully with prediction correctness. When the predictive entropy exceeds the threshold τ, the system activates a human-in-the-loop protocol, either requesting the learner to rewrite the character more clearly or deferring the assessment to a subsequent review session. This design philosophy—that it is better to admit uncertainty than to commit confidently to an error—aligns with established principles of responsible AI deployment.

B. Analysis of the Generalization Gap

The approximately 10% gap between training and validation accuracy warrants careful interpretation. While this gap might superficially suggest overfitting, several factors argue against this diagnosis:

1) The validation loss remains stable throughout training (Fig. 3) and does not exhibit the characteristic upward trajectory associated with overfitting.
2) The gap is consistent with the intrinsic difficulty of Kanji recognition, where thousands of characters share common radical components (e.g., the characters for "wait" and "attend" differ by a single stroke).
3) Label smoothing and dropout regularization provide effective constraints against memorization of training-specific patterns.

The mild oscillation observed in validation accuracy during epochs 10 through 13 (Fig. 2) suggests that the model is navigating a relatively flat region of the loss landscape, where small parameter perturbations can shift predictions on boundary cases. We hypothesize that collecting additional training data from underrepresented demographic groups—such as left-handed writers, whose stroke dynamics differ systematically from the right-handed majority—would improve generalization and reduce this oscillation.

C. Computational Cost Considerations

The 100+ hour training investment, while substantial, represents a one-time cost that is amortized over the lifetime of the deployed system. Once trained, the model operates at real-time speeds on consumer-grade hardware without requiring GPU acceleration for single-pass inference. The MC Dropout mode, which does require additional computation, can be selectively activated only when high-confidence predictions are needed (e.g., for formal assessment) while defaulting to single-pass mode for casual practice sessions.

D. Limitations

Several limitations of the current system should be acknowledged. First, the system processes characters individually and does not address the problem of segmenting connected or overlapping characters within a continuous text line. Second, the attention-based localization network assumes a single dominant character per frame, which precludes multi-character simultaneous recognition. Third, the current evaluation is conducted primarily on controlled indoor environments; performance under extreme conditions (e.g., outdoor lighting, colored paper) has not been systematically assessed. Finally, the model's

performance on rare or archaic Kanji characters, which are underrepresented in the training data, may be suboptimal.

VI. Conclusion

This paper has presented the design, implementation, and evaluation of a real-time Japanese handwriting recognition system that comprehensively addresses all three Japanese scripts: Hiragana, Katakana, and Kanji. Through the integration of Strong-CLAHE preprocessing, attention-based character localization, and a Bayesian EfficientNet architecture with Monte Carlo Dropout inference, the proposed pipeline achieves 99.8% training accuracy and 89.2% validation accuracy while maintaining real-time throughput on standard hardware.

The incorporation of principled uncertainty quantification represents a significant advance over conventional recognition systems, enabling the deployed educational application to provide calibrated, trustworthy feedback. The top-k scoring mechanism further enhances the learning experience by adopting an encouraging rather than punitive assessment philosophy.

Future work will pursue three complementary directions. First, we plan to integrate reinforcement learning from human feedback (RLHF) to enable continuous, lightweight model adaptation based on user interactions, progressively personalizing the recognition model to individual handwriting styles. Second, we will extend the pipeline to support multi-character recognition from continuous text lines, enabling more naturalistic writing assessment. Third, we intend to investigate federated learning approaches that allow model improvement across distributed user bases while preserving data privacy.

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